

1. Introduction

Strong lensing by galaxy clusters provides one of the most direct constraints on the projected mass distribution in dense environments (Bartelmann & Schneider 2001). Modern parametric reconstructions represent the cluster mass as the superposition of a few large-scale halos and many galaxy-scale subhalos, allowing precise recovery of critical curves and multiple-image configurations (Kneib et al. 1996; Jullo et al. 2007).

A standard choice for the galaxy-scale component is the pseudo-isothermal elliptical mass distribution (PIEMD, or dPIE in the spherical limit), often with vanishing core radius for cluster members (Jullo et al. 2007; Bergamini et al. 2019). In this formalism, galaxy parameters are linked through luminosity-based scaling laws, reducing the dimensionality of the model and enabling efficient optimization against large sets of lensing constraints.

The main limitation is interpretability: PIEMD parameters are convenient for lens modeling, but they are not directly equivalent to physically motivated stellar and dark-matter decompositions. In particular, concentrations and virial masses are naturally defined for NFW-like halos (Navarro et al. 1997), while stellar masses are more naturally tied to centrally concentrated baryonic profiles.

This work addresses that gap. Using the local analysis pipeline in `fit_bergamini_bmo_free_jaffe_grid.py` and `tutorial_density_profile_fits.ipynb`, we decompose PIEMD galaxy profiles into a Jaffe stellar component plus a truncated NFW-like dark-matter component (BMO form), then derive the implied c_{200} – M_{200} relations for four clusters. The paper emphasizes methodology and reproducible quantitative outputs from these scripts.

2. Methodology

2.1. Modeling of cluster galaxies

Cluster strong-lensing models are built as a sum of:

1. large-scale cluster halos, and
2. a population of galaxy-scale halos associated with member galaxies.

For the galaxy component, we adopt spherical PIEMD profiles with $r_{\text{core}} \rightarrow 0$ (the choice used in the fitting workflow; Jullo et al. 2007; Bergamini et al. 2019):

$$\rho_{\text{PIEMD}}(r) = \frac{\sigma_0^2}{2\pi G} \frac{r_{\text{cut}}^2}{r^2(r^2 + r_{\text{cut}}^2)}. \quad (1)$$

For completeness, the finite-core expression is

$$\rho_{\text{PIEMD}}(r) = \frac{\sigma_0^2}{2\pi G} \frac{r_{\text{cut}} + r_{\text{core}}}{r_{\text{core}}^2 r_{\text{cut}}} \frac{1}{(1 + r^2/r_{\text{core}}^2)(1 + r^2/r_{\text{cut}}^2)}. \quad (2)$$

In parametric lensing reconstructions, σ_0 and r_{cut} are usually tied to galaxy luminosity L through power-law scalings:

$$\sigma_0 = \sigma_{0,\star} \left(\frac{L}{L_\star} \right)^\alpha, \quad r_{\text{cut}} = r_{\text{cut},\star} \left(\frac{L}{L_\star} \right)^\delta. \quad (3)$$

Eliminating L gives

$$r_{\text{cut}} = r_{\text{cut},\star} \left(\frac{\sigma_0}{\sigma_{0,\star}} \right)^\beta, \quad \beta = \delta/\alpha. \quad (4)$$

This is the relation used and measured in our workflow.

2.2. Decomposing the PIEMD

We model each PIEMD profile as

$$\rho_{\text{PIEMD}}(r) \simeq \rho_{\text{Jaffe}}(r; M_\star, r_J) + \rho_{\text{BMO}}(r; \rho_s, r_s, r_t), \quad (5)$$

where the stellar component is (Jaffe 1983)

$$\rho_{\text{Jaffe}}(r) = \frac{M_\star r_J}{4\pi r^2(r + r_J)^2}, \quad (6)$$

and the dark-matter component is a smoothly truncated NFW profile in BMO form (Baltz et al. 2009), based on the NFW family (Navarro et al. 1996, 1997):

$$\rho_{\text{BMO}}(r) = \frac{\rho_s}{x(1+x)^2} \left(\frac{\tau^2}{x^2 + \tau^2} \right)^{1/2}, \quad x \equiv r/r_s, \tau \equiv r_t/r_s. \quad (7)$$

With $n = 1/2$ in Eq. 7, the asymptotic slope is close to r^{-4} , consistent with the large-radius PIEMD truncation behavior.

The fits are performed in log-density space over each galaxy radial grid, solving for $(r_s, r_t, \rho_s, M_\star, r_J)$. The quality metrics saved by the pipeline are $\text{RMSE}_{\log \rho}$ and $R_{\log \rho}^2$. For the four cluster runs used here, the median values are:

$$\text{median}(\text{RMSE}_{\log \rho}) \approx 0.045 \text{ dex}, \quad \text{median}(R_{\log \rho}^2) \approx 0.99998, \quad (8)$$

with small cluster-to-cluster scatter.

Figure 1 shows a 3×3 subset of PIEMD profiles and best-fit Jaffe+BMO decompositions.

2.3. Data

We analyze reconstructions for MACSJ0416, MACSJ1206, AS1063, and A2744, following the Bergamini et al. model family (Bergamini et al. 2019, 2023a,b). Redshifts and component counts are read from the local `LensTool` .par files used by the scripts. Multiple-image and source-family counts are taken from the corresponding published model papers.

Table 1 summarizes the key properties used in this work.

3. Results

3.1. Concentration-Mass relation

From each best-fit BMO halo, we compute R_{200} , M_{200} , and $c_{200} = R_{200}/r_s$. The derived concentration-mass trends are cluster dependent and very tight in each run. The median values are:

- MACSJ0416: $\tilde{c}_{200} = 22.75$, $\tilde{M}_{200} = 2.34 \times 10^{11} M_\odot$,
- MACSJ1206: $\tilde{c}_{200} = 61.37$, $\tilde{M}_{200} = 9.28 \times 10^{10} M_\odot$,
- AS1063: $\tilde{c}_{200} = 49.14$, $\tilde{M}_{200} = 1.18 \times 10^{11} M_\odot$,
- A2744: $\tilde{c}_{200} = 19.32$, $\tilde{M}_{200} = 2.90 \times 10^{11} M_\odot$.

The fitted log-slopes of the measured relations, $\log_{10} c_{200} = a + b \log_{10}(M_{200}/10^{12} M_\odot)$, are listed in Table 2 and span from $b \approx -0.23$ (AS1063) to $b \approx 0$ (A2744).

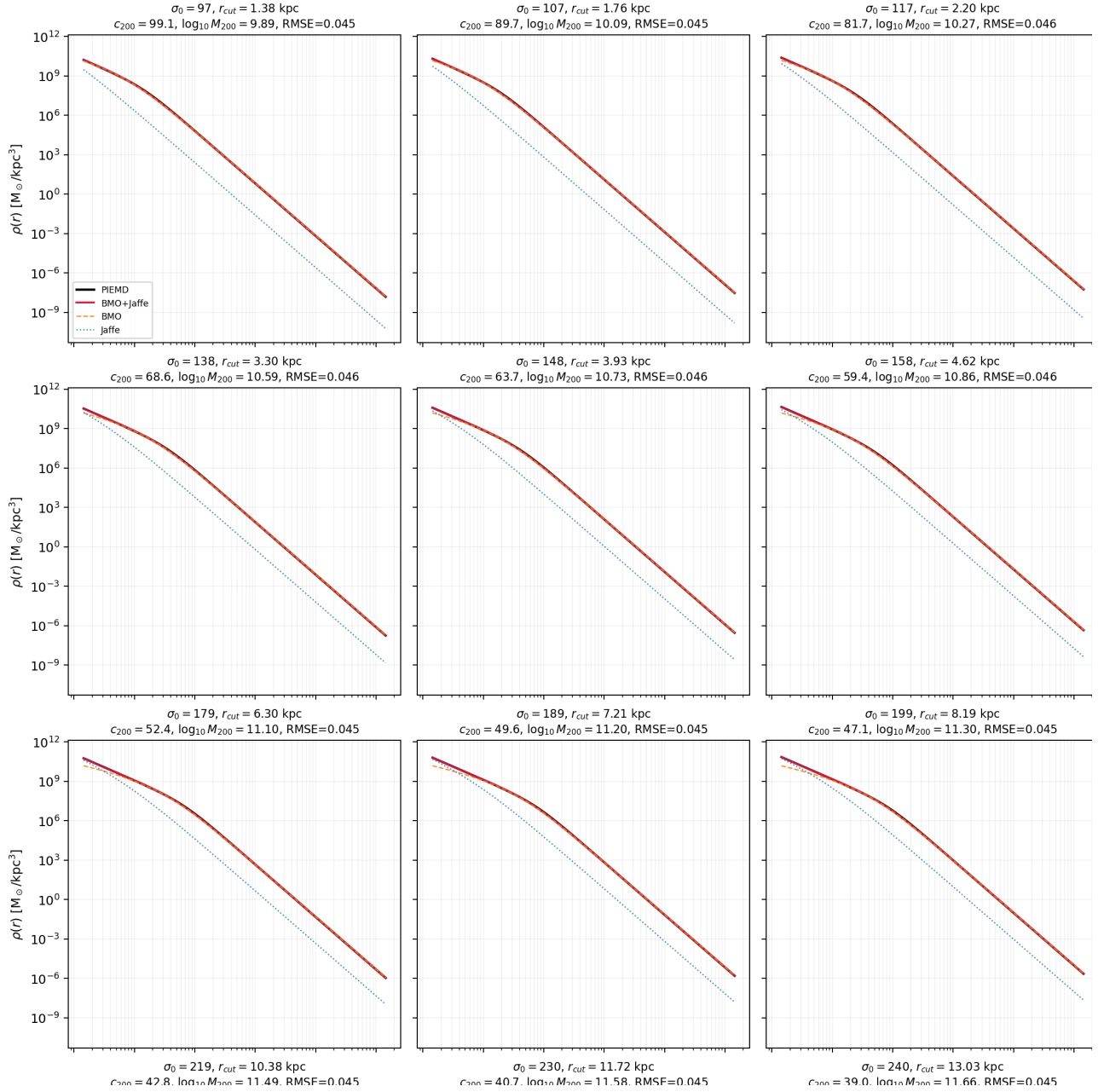


Fig. 1. Examples of PIEMD profile decompositions into Jaffe (stellar) + BMO (dark matter), extracted from the fit products of `fit_bergamini_bmo_free_jaffe_grid.py`. Each panel reports σ_0 , r_{cut} , best-fit c_{200} , M_{200} , and log-space fit RMSE.

3.2. Parametric modeling of galaxy concentrations

Figure 2 shows the measured σ_0 – r_{cut} relations. For each cluster we fit

$$r_{\text{cut}} = r_{\text{cut,ref}} \left(\frac{\sigma_0}{220 \text{ km s}^{-1}} \right)^\beta. \quad (9)$$

We then map $(\sigma_0, \beta, r_{\text{cut,ref}}, z)$ into (M_{200}, c_{200}) using the two-stage empirical model implemented in `tutorial_density_profile_fits.ipynb`. Defining

$$x \equiv \log_{10} \left(\frac{\sigma_0}{220} \right), \quad u \equiv \log_{10} \left(\frac{r_{\text{cut,ref}}}{10 \text{ kpc}} \right), \quad v \equiv \beta, \quad (10)$$

we write

$$\log_{10} \left(\frac{M_{200}}{10^{12} M_\odot} \right) = a_M + b_M x, \quad \log_{10}(c_{200}) = a_c + b_c x, \quad (11)$$

with

$$a_M = -0.5296 + 0.9376 u + 0.0020 v, \quad (12)$$

$$b_M = 2.1215 - 0.1385 u + 0.9502 v, \quad (13)$$

$$a_c = 1.6437 - 0.6875 u - 0.00037 v, \quad (14)$$

$$b_c = 0.6894 + 0.00789 u - 0.6941 v. \quad (15)$$

To account for redshift dependence in the conversion between BMO parameters and (M_{200}, c_{200}) , we define

$$q(z) \equiv \log_{10} \left[\frac{\rho_{\text{crit}}(z)}{\rho_{\text{crit}}(z_{\text{ref}})} \right], \quad z_{\text{ref}} = 0.396, \quad (16)$$

consistent with the expected role of evolving reference density and pseudo-evolution in halo-based quantities (Diemer et al.

Table 1. Cluster sample and model characteristics used in this analysis.

Cluster	z_{lens}	Main data sets	N_{LS}	N_{gal}	N_{mult}	N_{src}	Reference
MACSJ0416	0.3960	HST imaging + VLT/MUSE spectroscopy	10	212	237	88	Bergamini et al. 2023a
MACSJ1206	0.4390	HST imaging + VLT/MUSE spectroscopy	7	258	82	27	Bergamini et al. 2019
AS1063	0.3480	HST imaging + VLT/MUSE spectroscopy	5	222	55	20	Bergamini et al. 2019
A2744	0.3072	HST/JWST imaging + spectroscopy	9	171	149	50	Bergamini et al. 2023b

N_{LS} and N_{gal} are counted from the local .par realizations (components without and with mag, respectively). Multiple-image/source counts are literature values for the corresponding published model generation.

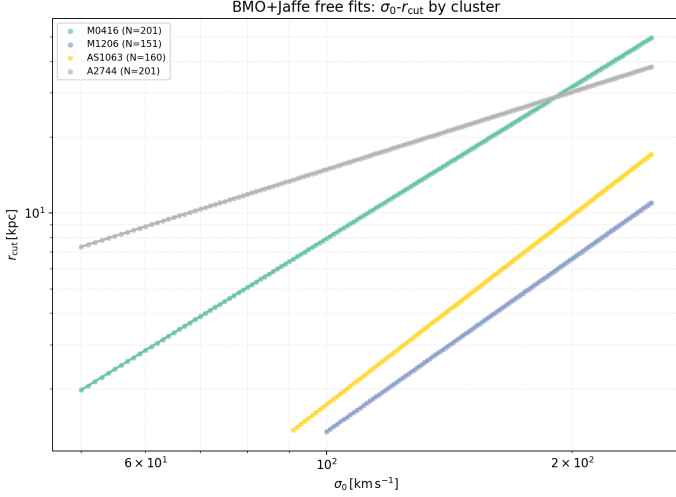


Fig. 2. σ_0 – r_{cut} relations for the four clusters, from `tutorial_density_profile_fits.ipynb`. Different $(r_{\text{cut,ref}}, \beta)$ pairs map into different c_{200} – M_{200} families.

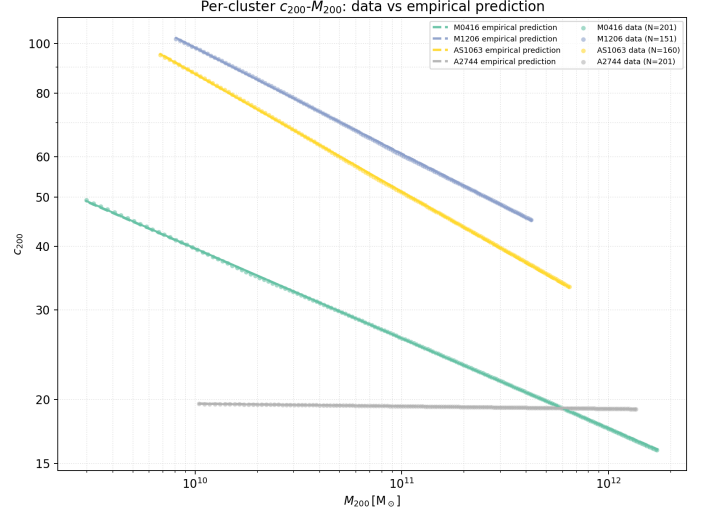


Fig. 3. Measured and empirical-predicted c_{200} – M_{200} relations for each cluster (data-driven calibration only).

2013; Diemer & Kravtsov 2015). and apply

$$\log_{10} \left(\frac{M_{200}}{10^{12} M_{\odot}} \right) = a_M + b_M x + \gamma_M q(z), \quad (17)$$

$$\log_{10}(c_{200}) = a_c + b_c x + \gamma_C q(z). \quad (18)$$

The fitted redshift coefficients in the notebook calibration are

$$\gamma_M = -0.0333, \quad \gamma_C = -0.3444. \quad (19)$$

The resulting predicted c_{200} – M_{200} relations match the measured ones very well (global residuals $\text{RMS}(\log M) \simeq 0.0034$ dex and $\text{RMS}(\log c) \simeq 0.0013$ dex for the calibration set). Figure 3 shows the agreement.

For external validation on previously unused clusters (PLCK-G287 and El Gordo), adding the redshift term reduces the median concentration residual from 0.0450 dex (no- z model) to 0.0066 dex (redshift-aware model). In El Gordo specifically, the concentration mismatch decreases from $\text{RMS}(\Delta \log c) \simeq 0.0866$ to $\simeq 0.0112$ dex.

4. Discussion

4.1. Overconcentration of cluster galaxies

Figure 4 compares the empirical cluster relations with standard LCDM concentration-mass prescriptions (Duffy et al. 2008; Dutton & Macciò 2014). At the median masses of our samples, observed concentrations exceed these expectations by factors of about 2.3–9.6 depending on the cluster.

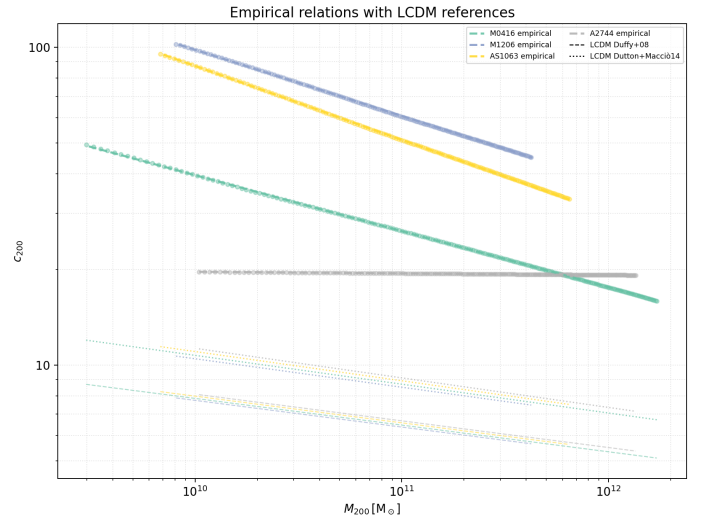
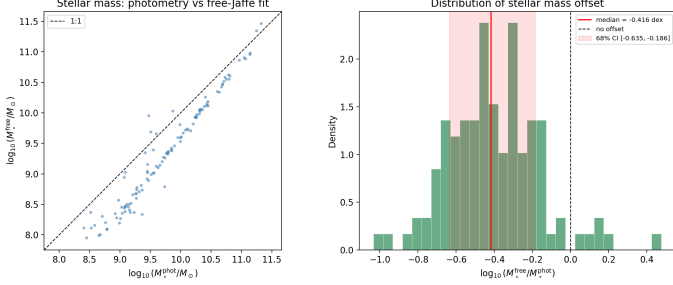


Fig. 4. Empirical c_{200} – M_{200} relations compared to LCDM benchmark trends (Duffy et al.; Dutton & Macciò). The observed relations are systematically shifted to higher concentrations at fixed mass.

The offset is largest in MACSJ1206 and AS1063, and smaller (but still significant) in MACSJ0416 and A2744. This cluster-to-cluster variation likely reflects a combination of genuine environmental differences and modeling systematics (e.g., parametric form rigidity, selection of subhalos strongly constrained by lensing, and central-region weighting).

Table 2. Empirical relation parameters from the cluster-by-cluster fits.

Cluster	N	$r_{\text{cut,ref}}$ [kpc]	β	$d \log c / d \log M$	$\tilde{c}_{200}$	$\tilde{M}_{200} [M_{\odot}]$
MACSJ0416	201	38.40	2.000	-0.1768	22.75	2.34×10^{11}
MACSJ1206	151	8.20	2.281	-0.2089	61.37	9.28×10^{10}
AS1063	160	12.44	2.493	-0.2312	49.14	1.18×10^{11}
A2744	201	33.40	1.022	-0.0049	19.32	2.90×10^{11}


Fig. 5. Stellar-mass comparison from Sect. 9.1 of `tutorial_density_profile_fits.ipynb`: photometric estimate versus free-Jaffe fit (left), and distribution of $\log_{10}(M_{\star}^{\text{free}}/M_{\star}^{\text{phot}})$ (right).

4.2. Inconsistency with stellar masses

The Jaffe component provides a lensing-based estimate of the stellar normalization. For MACSJ0416, comparing the fitted stellar normalization with the photometric estimate (derived from the F160W-based calibration used in the pipeline, in the spirit of Grillo et al. 2015) yields a median ratio

$$\text{median}(M_{\star,\text{lens}}/M_{\star,\text{phot}}) \simeq 1.96, \quad (20)$$

with a broad distribution.

This mismatch indicates that either:

1. the PIEMD-based decomposition is forcing part of the central dark matter into the stellar term,
2. the assumed stellar profile family (Jaffe) is not fully capturing real inner structure, and/or
3. the photometric stellar-mass calibration is biased for these cluster members.

Figure 5 shows the global comparison from the BMO+Jaffe dual-fit outputs.

5. Summary and conclusions

We developed and applied a reproducible workflow to translate PIEMD cluster-member models into a physically interpretable Jaffe+BMO decomposition, and to derive the implied c_{200} – M_{200} relation for galaxy-scale halos in cluster cores.

Main results:

- PIEMD profiles are fitted by Jaffe+BMO with excellent accuracy (median $\text{RMSE}_{\log \rho} \approx 0.045$ dex, median $R^2_{\log \rho} \approx 0.99998$).
- Cluster-specific σ_0 – r_{cut} relations differ substantially, with $(r_{\text{cut,ref}}, \beta)$ varying from (8.2, 2.28) in MACSJ1206 to (38.4, 2.00) in MACSJ0416.
- These differences map directly into distinct c_{200} – M_{200} relations.

- The empirical parameterization implemented in `tutorial_density_profile_fits.ipynb` reproduces the measured relations with residuals at the 10^{-3} dex level for this dataset.
- A redshift-aware correction based on $\rho_{\text{crit}}(z)$ is required for robust transfer to higher- z clusters; it strongly improves external validation (notably El Gordo).
- In all clusters, inferred concentrations are significantly above standard Λ CDM expectations at fixed mass.
- Stellar masses inferred from the lensing decomposition can disagree with photometric estimates, highlighting a key systematic that must be controlled in future analyses.

Future work should include a hierarchical model combining all clusters simultaneously, explicit treatment of covariance between scaling-law parameters, and tests with alternative stellar and dark-matter profile families.

References

- Baltz, E. A., Marshall, P., & Oguri, M. 2009, JCAP, 01, 015
 Bartelmann, M., & Schneider, P. 2001, Phys. Rep., 340, 291
 Bergamini, P., Rosati, P., Grillo, C., et al. 2019, A&A, 631, A130
 Bergamini, P., Grillo, C., Rosati, P., et al. 2023a, A&A, 674, A79
 Bergamini, P., Acebron, A., Grillo, C., et al. 2023b, ApJ, 952, 84
 Diemer, B., More, S., & Kravtsov, A. V. 2013, ApJ, 766, 25
 Diemer, B., & Kravtsov, A. V. 2015, ApJ, 799, 108
 Duffy, A. R., Schaye, J., Kay, S. T., & Dalla Vecchia, C. 2008, MNRAS, 390, L64
 Dutton, A. A., & Macciò, A. V. 2014, MNRAS, 441, 3359
 Grillo, C., Karman, W., Suyu, S. H., et al. 2015, ApJ, 800, 38
 Jaffe, W. 1983, MNRAS, 202, 995
 Jullo, E., Kneib, J.-P., Limousin, M., et al. 2007, New J. Phys., 9, 447
 Kneib, J.-P., Ellis, R. S., Smail, I., Couch, W. J., & Sharples, R. M. 1996, ApJ, 471, 643
 Navarro, J. F., Frenk, C. S., & White, S. D. M. 1996, ApJ, 462, 563
 Navarro, J. F., Frenk, C. S., & White, S. D. M. 1997, ApJ, 490, 493